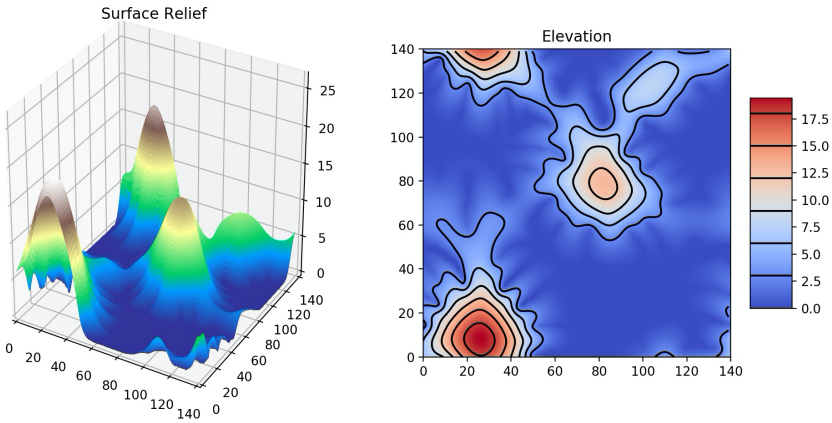
Homework 9: Erosion of a Landscape That You Created

For this homework, you will use an existing landscape evolution code that differs from the computer lab's version by the presence of an ocean and the absence of uplifting forces. In the first part of the assignment you will be asked to interpret the provided simulation code, and in the second part, you will be asked to simulation the erosion of a landscape that you created yourself.



(1) Download the Python file, *landscapeWithOcean.py* and the notebook *erosion_with_ocean.ipynb*. Execute all cells in the notebook. (If the code runs too slowly on your computer, reduce the number of grid points NX or NY.) Describe in your own words, how the initial conditions are chosen and what geological features they represent.

Have a look inside *landscapeWithOcean.py* and give an interpretation of the variable `'oceanLevelParameter'`.

Run the code several times and describe the resulting evolution. Say how the geological features change with time. What happens in the ocean?

(2) Please modify the code and replace existing initial conditions with your own ideas and code. *(Please do not use random numbers anywhere because we may not be able to grade your homework properly.)* Creating your own landscape may take some time. Part of the homework grade will be given for ***creativity.*** Describe what geological features you were trying to model even if their representation is not perfect.

Without changing the simulation parameters, erode your landscape with the provided code. Copy over the essential commands from *wave2d.ipynb* on bCourses, generate an .mp4 file of your landscape evolution and submit along with your notebook. Briefly describe how your landscape evolved and identify effects that are missing from your simulation if there are any.

(3) Now change the simulation parameters including K, D, n, m, the ocean level, and maybe dt and also the spatial resolution in order to erode you own landscape in a different way. Say what effects

the parameter changes had. Points be assigned for good descriptions. See if you find some interesting effects. Again, we give points for creativity. Submit your notebook and a second mp4 file that contains your best landscape evolution model.